

Computer Netzwerke und verteilte Systeme Summer 2026



Exercise 11: Application Layer

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Deadline: The deadline for this exercise is **07.07.2026, 23:55** and in total **9 bonus points** can be achieved.

Bonus points: *Computer Netzwerke und verteilte Systeme* features a bonus point system to acknowledge first-rate performance. Exercise points count as bonus. Throughout the entire course, a bonus of up to a maximum of 0.7 can be obtained. The bonus does not help in passing the exam: You need to pass the exam with a 4.0 or better before any bonus is applied.

Group size: The recommended group size for submissions is 3. You may also submit your exercises as a group of 2 or alone if you do not find a group.

Important: Every theoretical task and subtask require proper sources. Submissions without proper sources will automatically receive 0 points.

For most of the task the slides of the lecture will be enough. For **research** task, use the materials provided in the lecture, or the internet.

Refrain from copying and pasting your answers, as this is considered **plagiarism** and will have **consequences**, as will using AI tools to create your answers. Instead, explain your answers in your own words and cite your sources!

Write down your answers in a legibly and unambiguously way, either by creating formatted documents (LaTeX, Word, typst, etc.) or by handwriting (**ensure that your handwriting is legible**).

Refrain from using colors that create less contrast between the text and the background. For marking tasks you may use blue and orange.

Task 1: Group Exercise: Socket

Assume you want to implement a distributed system on top of the Socket API.

- (a) Which basic operations do exist in this API?
 - 1) For connection-less protocols
 - 2) For connection-oriented protocols
- (b) To which API does the socket API bear similarity?
- (c) Which information is needed to uniquely identify a connection using the Socket API?

Task 2: Group Exercise: Distributed Applications

Speaking about distributed applications:

- (a) What are distributed applications?
- (b) What are the pros and cons of distributed applications?
- (c) What are the differences between open and proprietary protocols?

Task 3: Group Exercise: HTTP

Speaking about the HTTP (hypertext transfer protocol):

- (a) Name three improvements of HTTP/2 in contrast to HTTP/1.0 and HTTP/1.1.
- (b) Describe the difference between non-persistent and persistent HTTP and name at least 3 differences. Which HTTP version is using non-persistent/persistent HTTP?
- (c) What is the meaning of the request message "GET /somedir/page.html HTTP/1.1" in the context of HTTP?

Task 4: Research: DNS (4.5 Points)

- (a) Imagine a fictional internet where DNS is hosted on a single, massive supercomputer in Frankfurt. What are two distinct operational risks of this architecture, and how does the actual, real-world distributed design of DNS solve them? (1.5 P)
- (b) Explain the difference between a recursive and an iterative DNS query. (1P)
- (c) A network administrator updates a website's IP address, but some users still see the old website for the next 24 hours. Identify the mechanism causing this behavior, explain how it benefits network performance, and state how the administrator could have prevented this delay. (2 P)

Attention: Submissions without references will not be graded! Please add sources.

Task 5: HTTP (4.5 Points)

- (a) Research: Why is "Head-of-Line blocking" a problem in HTTP/1.1 and how does HTTP/2 fix it? Briefly explain how HTTP/3 fixes it on the transport layer. (1.5P)
- (b) On the application layer, HTTP is fundamentally stateless. However, a modern e-commerce platform needs to track a single user's actions across multiple sequential clicks. Based on the architecture shown in the lecture slides, describe the exact step-by-step technical process of how a stateful session is established and maintained. In your answer, explicitly name the **two distinct HTTP header lines** involved and explain where the user's data record is actually stored. (1.5 P)
- (c) Research: What does the HTTP response code 429 indicate and what HTTP response header should the server send to the client with this response code? (1.5P)

Attention: Submissions without references will not be graded! Please add sources.